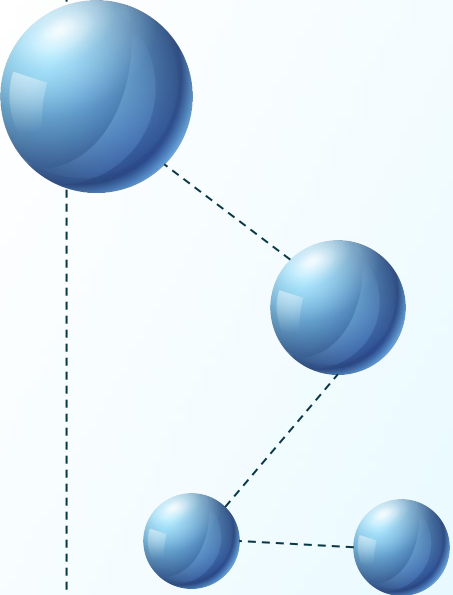
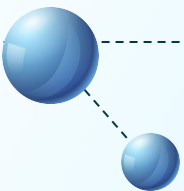
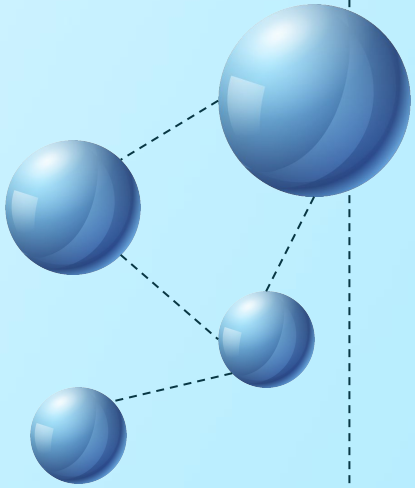
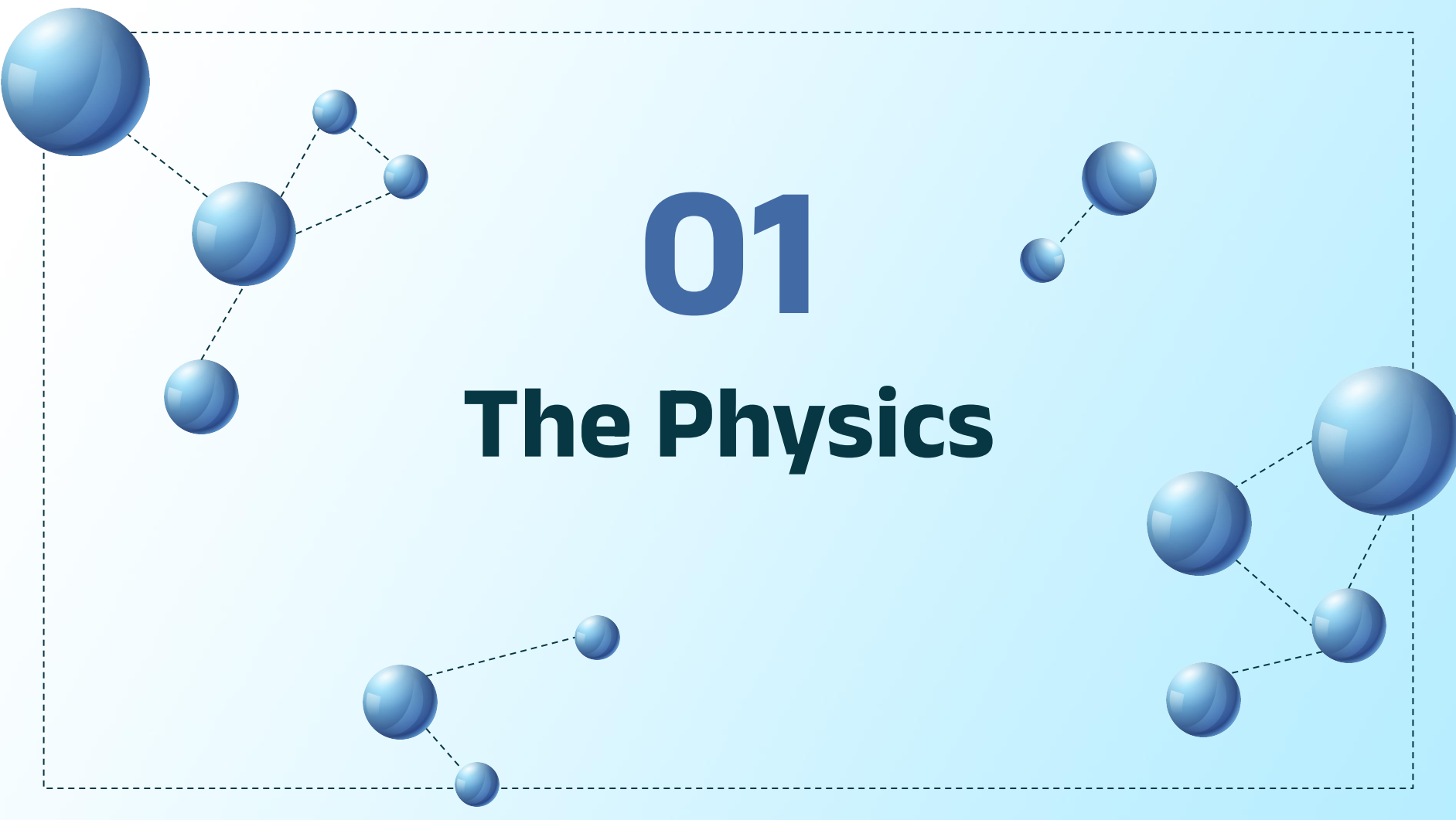




MRI BASIC PHYSICS

Field Crumley and Calvin Mar

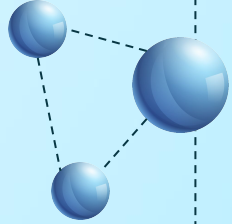


The slide features a light blue background with a dashed rectangular border. Scattered around the central text are several molecular models. Each model consists of blue spheres of varying sizes, representing atoms, connected by dashed lines representing chemical bonds. The models are positioned in the corners and along the sides of the slide, creating a scientific or chemical theme. The text '01' is large and blue, while 'The Physics' is in a smaller, bold, dark blue font.

01

The Physics

Absorption/Emission, photon



$$E^2 = (mc^2)^2 + (pc)^2$$

For a photon:

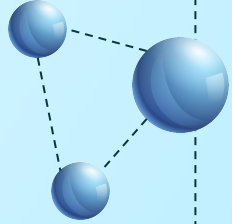
$$p = \frac{h\nu}{c}$$

Absorption/Emission rules:

$$E_0 + E_\lambda = E_1$$

$$\implies \Delta E = E_\lambda$$

Spin, Quantization of Energy



$$m_i = \pm \frac{1}{2}$$

Corresponding to these spins,

$$E_i = -m_i \left(\frac{\gamma h \vec{B}_0}{2\pi} \right)$$

Larmor Precession

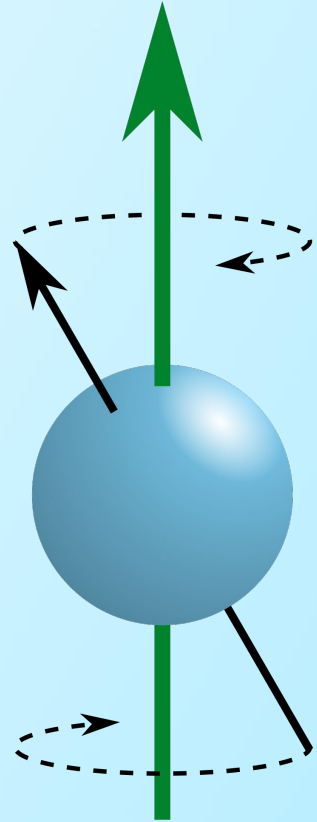
We need a B_0 to make $\nu_\lambda = \nu_H = \frac{\gamma B_0}{2\pi}$, so:

$$\omega_H = \gamma B_0$$

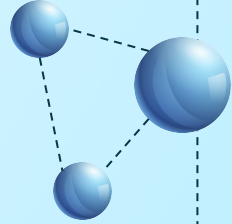
$$2\pi\nu_H = \gamma B_0$$

$$\nu_H = \frac{\gamma B_0}{2\pi}$$

$$B_0 = \frac{2\pi\nu_\lambda}{\gamma}$$



The Boltzmann Distribution



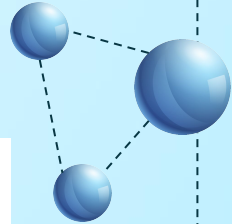
$$h\nu + E_0 \rightarrow E_1$$

Interestingly, $P(E_0 \rightarrow E_1) = P(E_1 \rightarrow E_0)$.
Solution?

$$P(E_n) \approx e^{-E_n/(kT)}$$

$$P(E_1)/P(E_0) = e^{E_0 - E_1/(kT)} < 1$$

Spin-Lattice Relaxation



leads to...

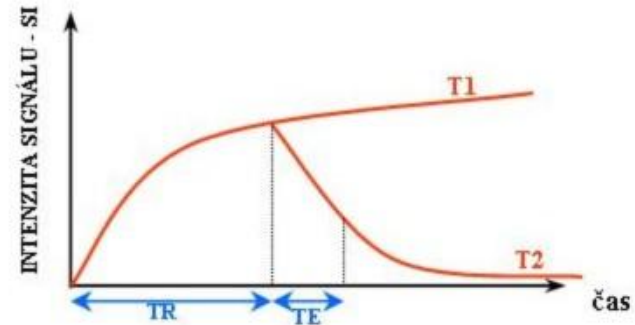
$$\frac{dM_z}{dt} = \frac{M_0 - M_z}{T_1}$$

$$M_z(t) = M_0(1 - e^{-t/T_1})$$

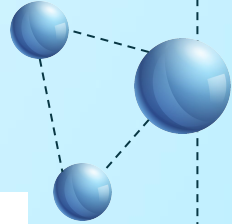
$(T_2 > T_1 \dots)$

Small $T_1 \implies$ bright,

Large $T_1 \implies$ dark.



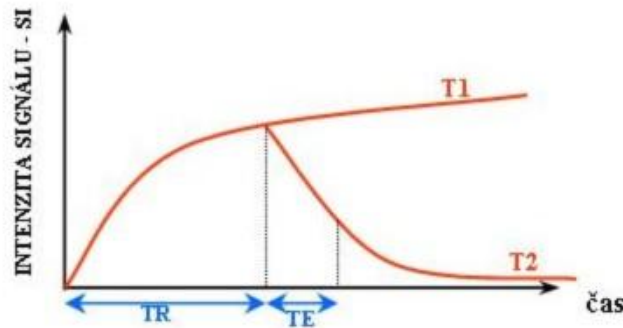
Spin-Spin Relaxation



We turn off the magnet, and:

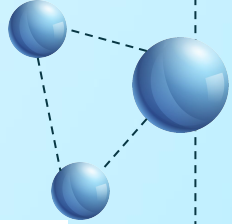
$$\frac{dM_z}{dt} = -\frac{M_1 - M_z}{T_2}$$

leading to...

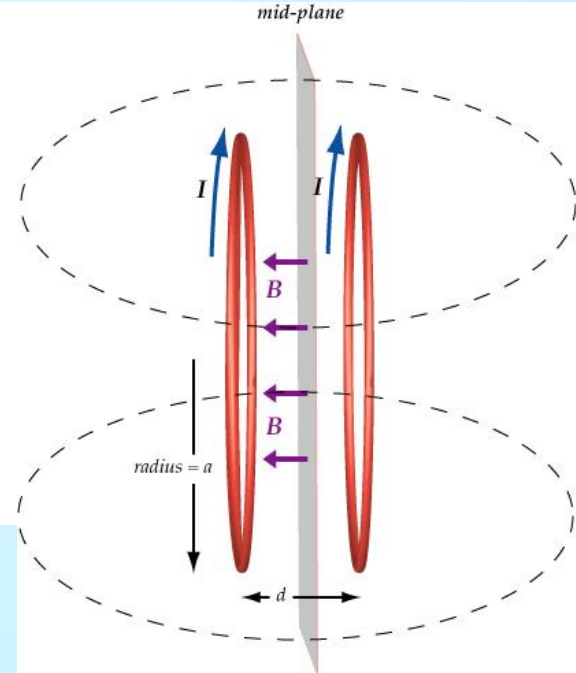


$$M_z(t) = M_1 - e^{-t/T_2}$$

The Biot-Savart Law



$$\vec{B} = \frac{\vec{v}}{c^2} \times \vec{E}$$



But... How to be Precise?

Recall the Larmor Angular Frequency:

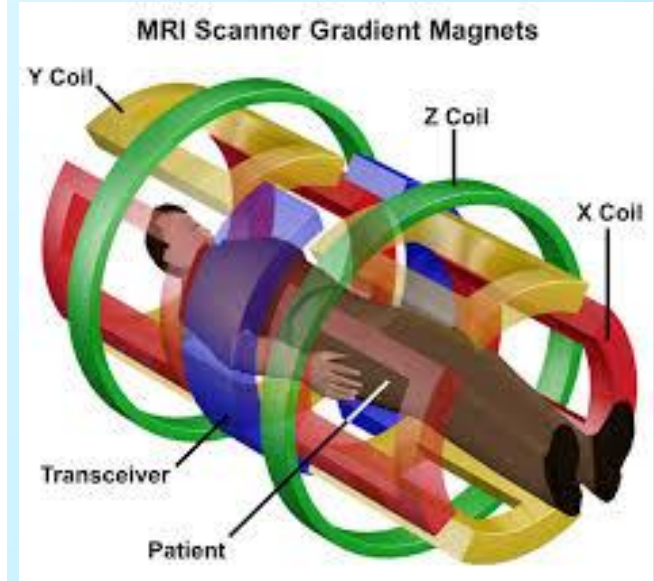
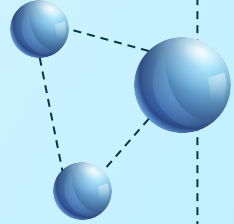
$$\nu_H = \frac{\gamma B_0}{2\pi}$$

$$\Rightarrow B_0 = \frac{2\pi\nu\lambda}{\gamma}$$

Now:

$$B = B_0 \rightarrow B(x) = G_x x$$

$$\Delta x = \frac{2\pi\nu\lambda}{\gamma G_x} = \frac{2\pi\Delta E}{\gamma G_x}$$



Recap

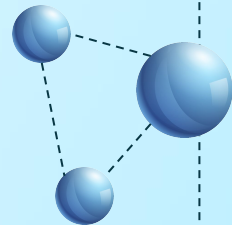
$$\vec{B} = \frac{\vec{v}}{c^2} \times \vec{E}$$

$$m_i = \pm \frac{1}{2}$$

$$E_i = -m_i \left(\frac{\gamma h \vec{B}_0}{2\pi} \right)$$

$$\omega_H = \gamma B_0$$

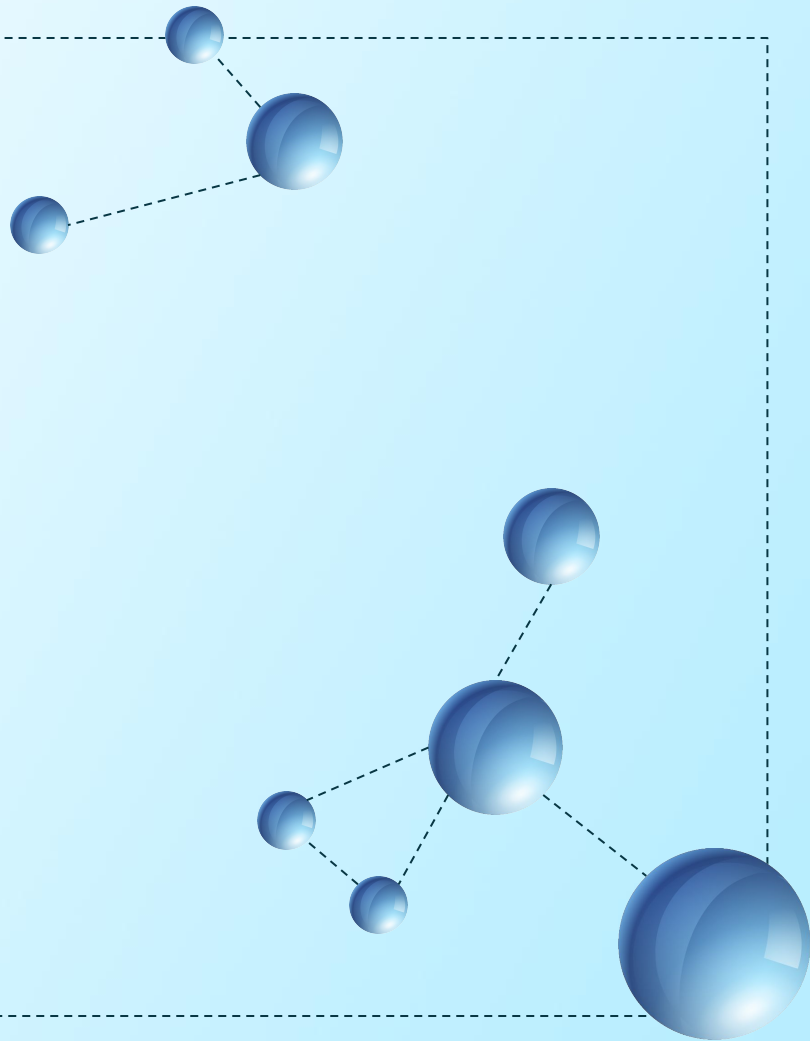
$$B(x) = G_x x$$



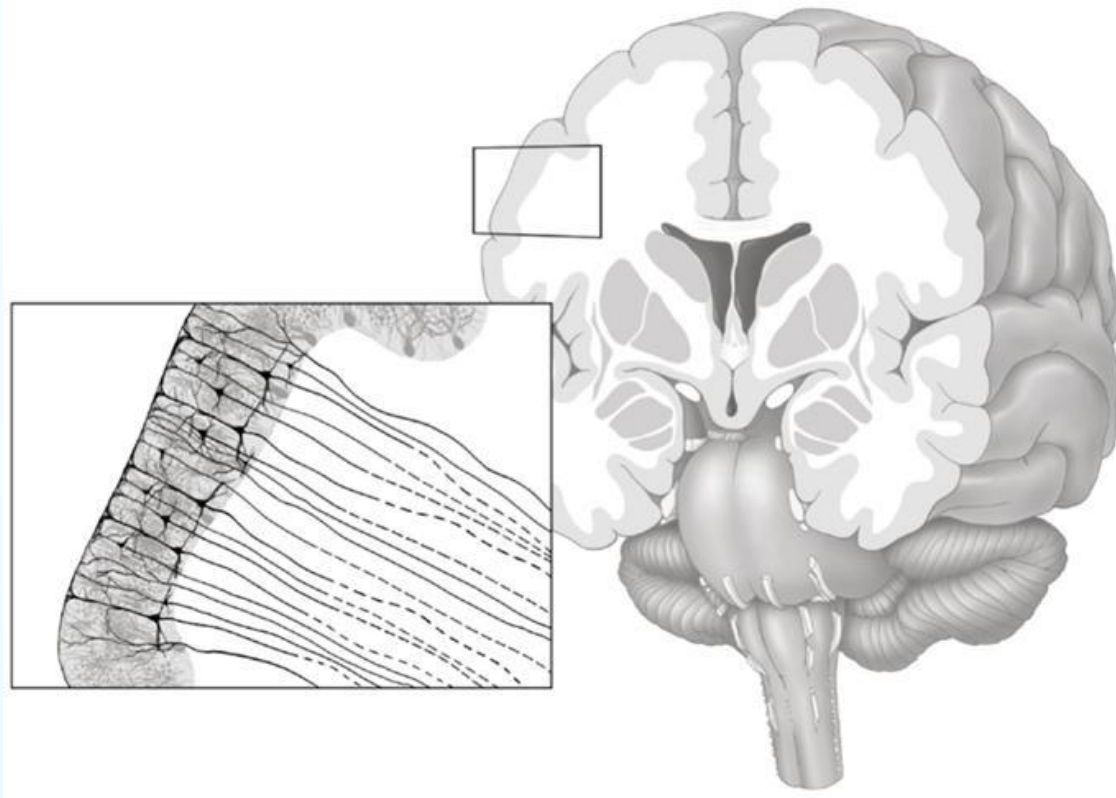
02

Composition of the Brain

What do those signals mean?

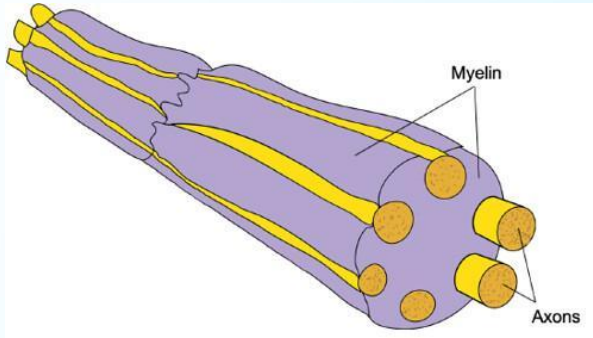


White Matter and Gray Matter



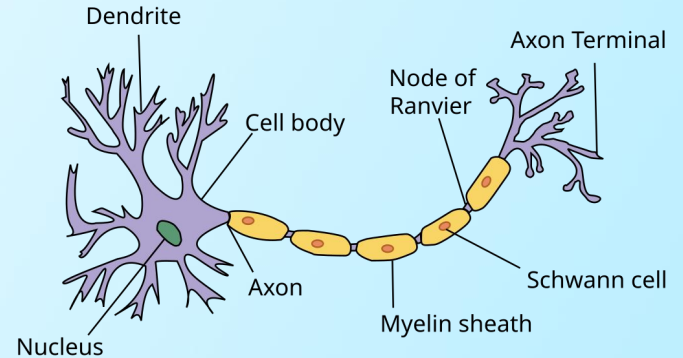
White Matter and Gray Matter

White Matter: Axons



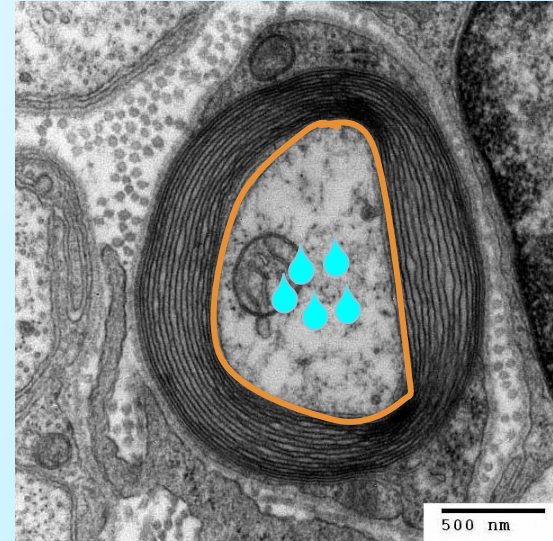
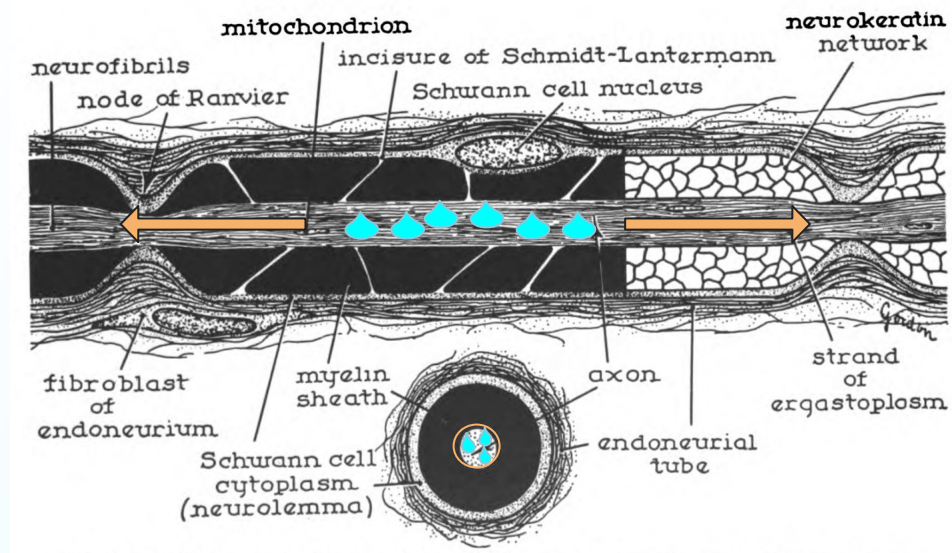
- Information carriers
- Bundled
- Water has directional freedom of movement (anisotropy)

Gray Matter: Neurons



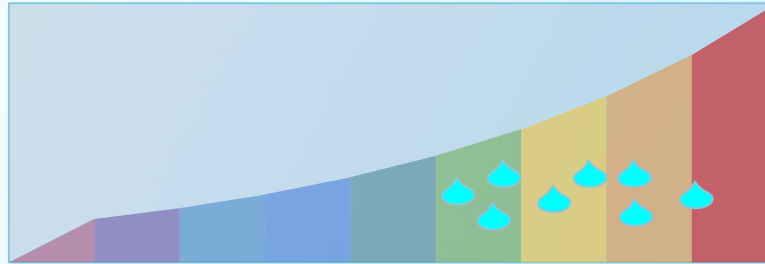
- Information storage
- Densely packed
- Water is relatively trapped in place (isotropy)

Diffusion of Water



Detecting Diffusion

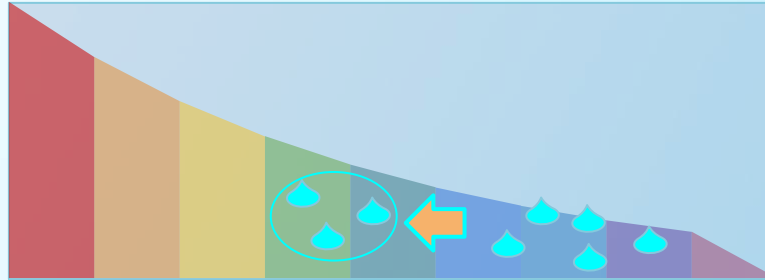
Homogeneous Field:
Locked Spin



Gradient Field:
Desynchronous Spin

Stejskal-Tanner
Equation

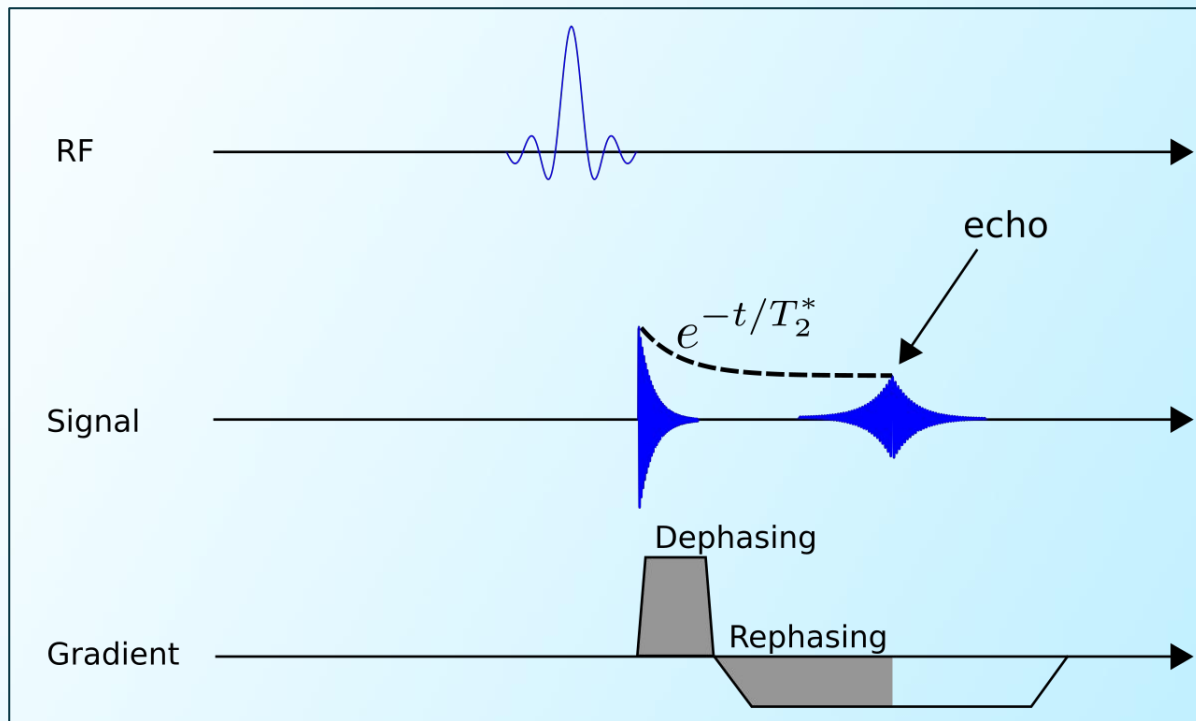
$$S = S_0 e^{-bD}$$

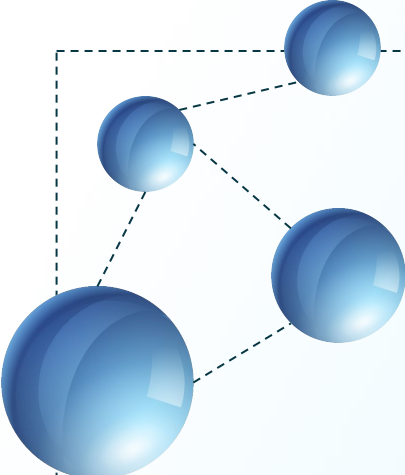


Gradient Field:
Resynchronize Spin

Homogeneous
Field: Signal Loss

MRI Gradient Echo Graph

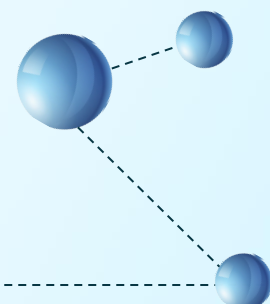




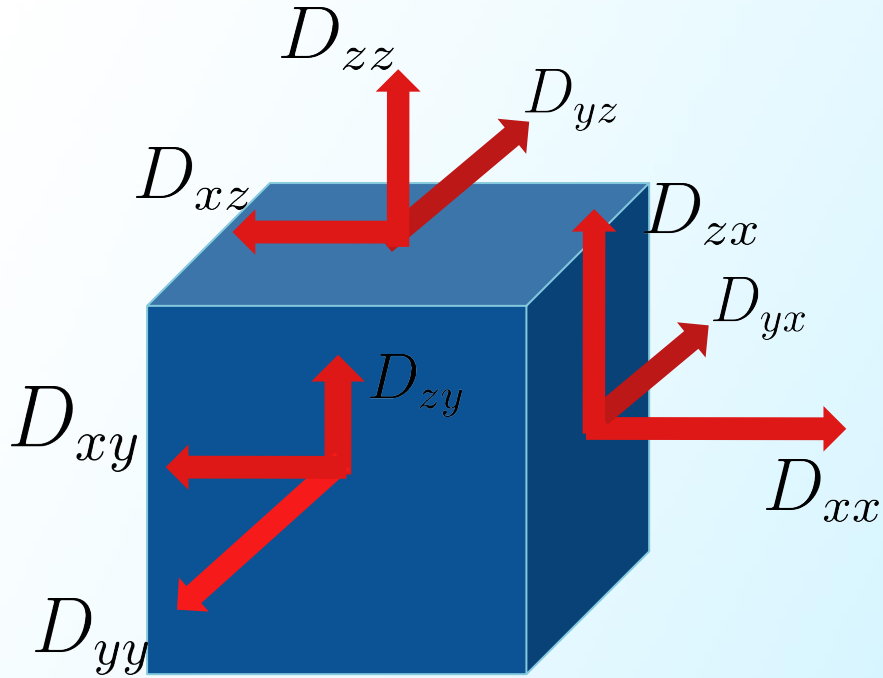
04

Constructing the Data

What's that number do?

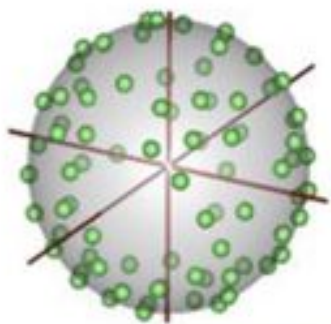


The Diffusion Tensor

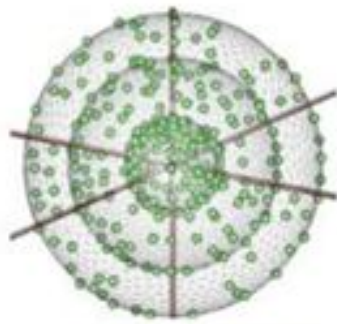


$$\begin{vmatrix} D_{xx} & D_{xy} & D_{xz} \\ D_{yx} & D_{yy} & D_{yz} \\ D_{zx} & D_{zy} & D_{zz} \end{vmatrix}$$

More Vectors!

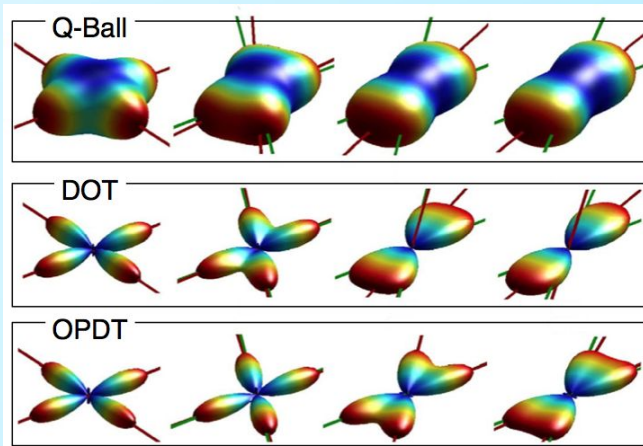
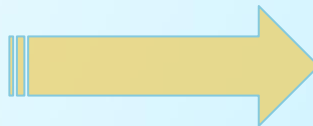


single-shell dMRI
(DTI, HARDI)



multi-shell dMRI
(3D diffusion function)

Funk-Radon



Thanks for Listening



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Resources

- <https://pmc.ncbi.nlm.nih.gov/articles/PMC3163395/>
- [https://www.news-medical.net/health/Diffusion-Tensor-Imaging-\(DTI\)-Explained.aspx](https://www.news-medical.net/health/Diffusion-Tensor-Imaging-(DTI)-Explained.aspx)
- <https://pmc.ncbi.nlm.nih.gov/articles/PMC3003887/>
- https://mri-q.com/uploads/3/4/5/7/34572113/tuch-2004_q-ball_-_magnetic_resonance_in_medicine.pdf
- <https://www.youtube.com/@MRIPhysicsEXPLAINED>
- <https://www.youtube.com/watch?v=nPaTG1f-JBU>
- <https://www.youtube.com/watch?v=jLnuPKhKXVM>
- https://en.wikipedia.org/wiki/Physics_of_magnetic_resonance_imaging
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- Macomber, R. S. (1998 - 1998). A complete introduction to modern NMR spectroscopy. Wiley, New York.
- Griffiths, D. J. (2013). Introduction to electrodynamics. Pearson.